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AI – DRIVEN BANDWIDTH OPTIMIZATION IN SOLAR – POWERED COMMUNITY MESH NETWORKS FOR RURAL CONNECTIVITY

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Content

Access to reliable internet is still a serious challenge for many rural communities. Poor infrastructure, high installation costs, and irregular electricity supply make it difficult to build and maintain traditional network systems in these areas. Because of this, important services like online education, telemedicine, digital banking, and government platforms often remain inaccessible to the very people who need them most.

To help bridge this gap, this work proposes a practical and sustainable solution that brings together two powerful ideas: solar-powered community mesh networks and intelligent bandwidth management using artificial intelligence.

In this approach, low-cost mesh network nodes are powered by solar energy, which means they do not depend on the main electrical grid. This makes them ideal for remote or off-grid regions where electricity is unreliable or unavailable. The mesh structure also allows communities to build flexible, self-healing networks that can expand easily as more nodes are added.

What makes this system unique is the addition of an AI-based bandwidth management system. Instead of treating all internet traffic the same, the system learns from real-time network conditions and user demand. It identifies which applications are most important — such as virtual classrooms, online medical

consultations, and government services — and gives them priority during times of congestion.

The AI model continuously monitors factors like network load, application urgency, and available solar power. Based on these conditions, it dynamically adjusts bandwidth allocation to ensure critical services remain smooth and responsive. This smart allocation not only improves user experience but also makes better use of limited network and energy resources.

Simulation results show that this integrated approach can reduce delays for essential services, improve overall network performance, and extend coverage while keeping energy consumption efficient. By combining renewable energy, community-driven networking, and intelligent traffic management, this solution offers a realistic and scalable path toward digital inclusion for underserved rural regions.